

Arpit Kumar Singh

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SUMMARY

Full-stack developer with hands-on experience building AI-native applications, including RAG pipelines, vector search, and LLM-integrated systems, alongside strong MERN and REST API skills. Solid foundation in DSA, system design, and scalable web architecture, with a genuine interest in agentic AI and the future of software engineering. Focused on shipping reliable, high-performance products end-to-end with clean code and measurable user impact.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL

Frontend: React.js, Next.js, HTML, CSS

Backend: Node.js, Express.js, REST APIs, JWT

Databases: MongoDB, MySQL, PostgreSQL

AI & LLM: RAG Pipelines, LLM Integration, Embeddings, Vector Databases (ChromaDB), Knowledge Graphs (Neo4j), Prompt Engineering, FastAPI, Sentence-Transformers

DevOps & Tools: Git, GitHub, CI/CD (GitHub Actions), VS Code

Concepts: Data Structures & Algorithms, System Design, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Software Engineer, RealityScale

Nov 2025 – Jan 2026

React, REST APIs, Authentication

- Built **reusable** React components, dynamic dashboards, and secure authentication flows, improving user experience while reducing frontend development time through a modular component architecture.
- Designed and **optimized** SQL database schemas and backend APIs, improving query performance.
- Collaborated with **cross-functional teams** to ship production features end-to-end, contributing to code reviews, debugging, and performance tuning across the stack.

PROJECTS

Industrial Knowledge AI — RAG + Knowledge Graph Platform

[Link](#)

Python, FastAPI, Spring Boot, React, ChromaDB, Neo4j, Google Gemini

- Built a **hybrid retrieval (RAG)** system fusing semantic vector search (ChromaDB, **all-MiniLM-L6-v2** embeddings) with a **Neo4j knowledge graph** to deliver source-grounded answers over heterogeneous industrial PDFs.
- Engineered an end-to-end ingestion pipeline: PDFBox text extraction, chunking (1000/200 overlap), embedding, and vector indexing, with **spaCy** entity extraction populating the knowledge graph for cross-document reasoning.
- Integrated **Google Gemini** for grounded answer generation constrained to retrieved context, decoupling the AI layer into a **FastAPI microservice** separate from the Java Spring Boot backend.
- Designed REST APIs across a three-tier architecture (React frontend, Spring Boot ingestion backend, Python AI service) for document upload and natural-language querying.

Hostel Complaint & Application Management System

[Link](#)

Node.js, Express.js, MongoDB

- Designed and developed a full-stack system to manage hostel complaints and administrative applications for a residential campus.
- Implemented **role-based access control** for students and administrators using secure authentication and session-based flows.
- Built a complete complaint lifecycle workflow with real-time status tracking (**Pending, In-Progress, Resolved**) and an admin dashboard to review complaints, view uploaded images, add remarks, and update resolution status.

POSITIONS OF RESPONSIBILITY

Joint Secretary, IEEE NSUT (The Largest Technical Society of NSUT)

Dec 2023 – Present

- Improved and managed the IEEE NSUT website for **TENSYMP'24**, enabling researchers to seamlessly **register and book presentation slots** through an optimized, user-friendly platform.
- Organized Web Development and AI/ML bootcamps, training **300+ students** on modern coding practices and project-based learning.
- Managed a **team of 40+** members, navigating problems and optimizing resources across multiple technical events.

Sponsorship Coordinator, Moksha Innovation 2024 (Annual Cultural Fest of NSUT)

Jan 2024 – Mar 2024

- Partnered with reputed companies including **SPARKS, DECATHLON, VISION IAS, and STORIA** to drive sponsor engagement.
- Successfully drove **customer acquisition** for the co-title sponsor, resulting in over **6K+ app downloads**.
- Managed back-end operations of the sponsorship department by formulating **MOUs** and negotiating deliverables with brand partners.

EDUCATION

Netaji Subhas University of Technology (NSUT), Delhi — B.Tech, Mechanical Engineering (CGPA: 7.0)

2023 – 2027

Tagore Academy — Class XII, ISC Board (74.8%)

2020 – 2022

Tagore Academy — Class X, ICSE Board (78.8%)

2019 – 2020

ACHIEVEMENTS

- Hackathon Winner:** Won a **24-hour hackathon** by designing and developing a full-stack AI chatbot for a coffee shop, enabling customers to browse products, receive personalized recommendations, and interact with an intelligent virtual assistant.
- 350+ DSA problems** solved in C++ across LeetCode, CodeForces, and CodeChef competitive programming platforms.
- Career Essentials in Software Development** — Microsoft & LinkedIn certified.
- Elite Certificate** — Design Thinking Course (IIT Madras).